

THE KERMOND CERTIFICATE OF EDUCATION

IP- Redemption Task 3

STUDENT NAME: _____

Teacher Code : HEWP XIAR **KERJ** SOTS FIRR **ALIY** **CHEW** **SRIK**
GOWS **BUYR** **PATV** **-RIBS**

Mathematical Methods 3/4

APPLICATION TASK 1B

Question and Answer Book

The Adventures of William Patel (tribute)



The Movie : JKSS World Cup

The Cup Final -Part II

Reading time is **15 Minutes**

Writing time is **120 Minutes**

Approved Materials:

- Any aids for the sketching of graphs
- A physical **OR** digital Mathematica Bound Reference
- **OR** A CAS Calculator with **physical** bound reference
- Stationary including pencils and a scientific calculator (if desired)

Materials supplied:

- Question and Answer book of k amount of pages

Contains

page

ANY PAGE BREAKS OR ERRORS ARE **INTENTIONAL**

6 Questions



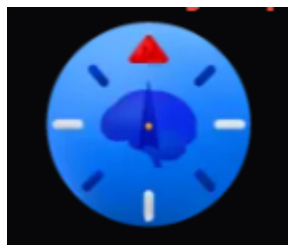
Flashback Part I:

The JKSS SAC I was stolen from Mr. Kermonds' office and William Patel an Y11 Methods $\frac{3}{4}$ Student, was assigned by Principal Adrian Chisholm to help Mr. Kermond recover the SAC, they had followed clues laid by Mr. Kermonds nemesis, the Schmiddler, and went to all sorts of places such as the Lucas Leap highway, the Path of Shawarma and Mr. Mays Circus

The gang was able to survive and meet each other after each trial and tribulation, a testimony to their resilience, after nearly being killed by a force field by an imaginary ghost, they returned to JKSS and decided to re-group operations from there. William Patel was told to engage in his studies and not worry too much about this debacle of the great SAC Robbery, however:

Suddenly Mr. Kermond received a phone call from schmiddler, it was a challenge- to the cup final of the JKSS World Cup

William knew something was wrong, he just didn't know how much of a magnitude, and direction it had ...



SECTION B:

- Answer **all** questions in the booklet
- All written responses need to be in English
- If not specified then provide **exact** values for each question
- Mathematica /CAS Syntax is **not** suitable as working out
- Show all mathematical reasoning if a question is of multiple marks
- The sections in italics are story only and not required for the assessment
- Use the diagram provided for you of the stadium in part I to help you answer the questions

Question 5 (9 Marks)

MISSION STATUS : STANDBY

TIME : 8:40 AM (GMT+10)

JKSS ZONE 2B3

William Patel is back in his Methods Class, As his teacher , *Mr. Kermond* is deciding to plan the team for the great cup final to beat The *Schmiddler's* team . He decides to test the football capabilities of various students in Will's class. He knows that is proportional to their ability

To do transformations

He writes a function on the board in the form of:

$$a(x) = \sqrt{10x - x^2}$$

He asks , *Stephanie* ,one of William's friends to state a sequence of transformations that map $a(x)$ to the graph of $9 - 3\sqrt{-3x^2 + 30x + 150}$

She applies the transformations in the wrong order and ends up with the graph of

$$1 - 9\sqrt{\frac{(50+10x-x^2)}{3}}$$

a. State the correct transformations , and explain the error(s) she made (3 Marks)

$$a_2(x) = 9 - \sqrt[3]{2x^2 + 18x + 9}$$

and

$$a3(x) = \sqrt[3]{x^2 - 27}$$

b. State a sequence of transformations that maps a_2 to the graph of a_3 (3 Marks)

Blank lined paper with a large diagonal watermark reading "AGD".

He asks the final suspect in William's friend group *Roshan* to do a new type of transformations question consider the function:

$$A: [k, \infty) \rightarrow R, A(x) = e^{2x} - 8e^x + 7$$

Where k is the smallest value so that the inverse is defined

C.

I. Show that using algebra that the rule of the inverse function is: (2 Marks)

$$\ln(4 + \sqrt{x + 9})$$

He says she may use this fact for the next question:

ii.) **Hence or Otherwise**, State **two** transformations she can use to transform $A(x)$ to the graph of

$$\ln\left(\frac{1}{36} (24 + \sqrt{324 + 36x})\right) \quad (2 \text{ Marks})$$

Mr Kermond has a good idea of who he should include for the football team for the cup final, "The Schmiddler has no chance against my troupe!!!"

Stephanie is a bit scared to play against the Schmiddler even though she is good at soccer and her mood, in a value expressing her intensity in mood, evolves according to the function

$$A_k(t) = e^{2t} - 8ke^t + 7 \text{ where } 0 < t < 67$$

Where t is the time in minutes after Mr. Kermond Selects her

d.)

i. For what value(s) of k does $A_k(t)$ intersect with the line $y = 6x + 7$ once (2 Marks)

ii. For what value(s) of k is her mood the worst after 67 seconds (1 Mark)

She will not be allowed to play if her mood **falls** below 20000 Mood units at half time

iii.) For what value(s) of k will she not be allowed to play?

At the beginning of the football match they encounter the setup described again

Question 6 (12 Marks)

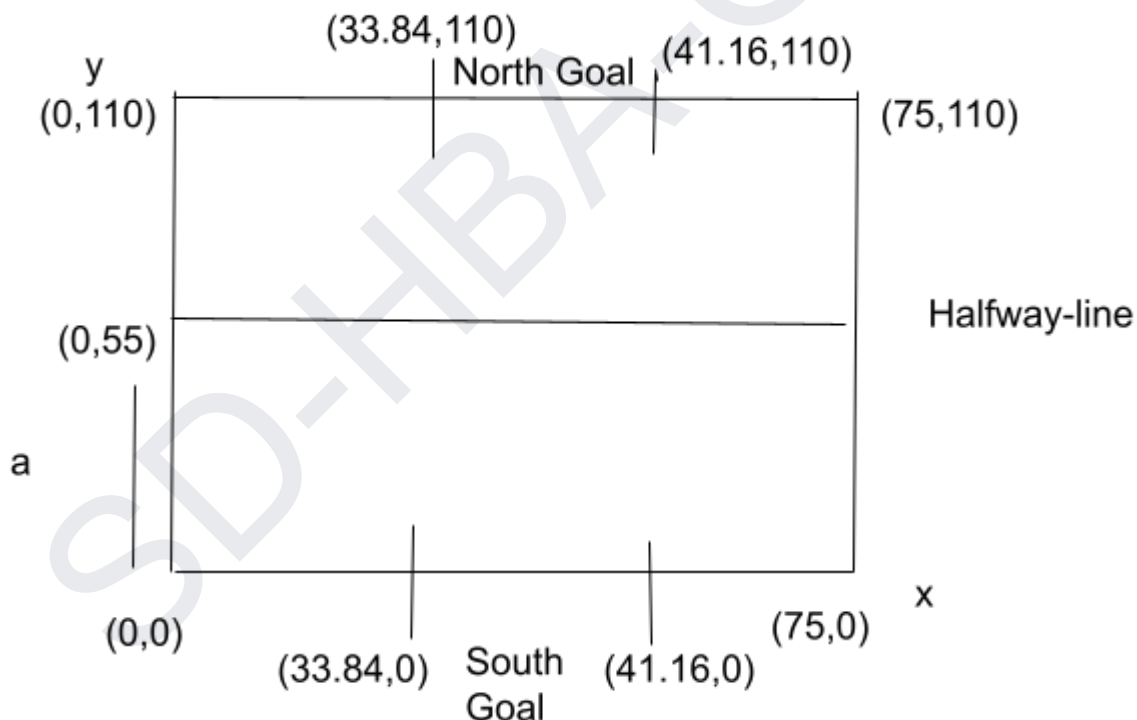
MISSION STATUS: ONTO SOMETHING MAYBE?

TIME: 9:40 GMT+11

LOCATION: The Grand Stadium

The JKSS football team , selected from the winners of each past game including William's patels three friends from methods.

JMSS Cup Final - All distances in metres (I think?!)



The following diagram shows a diagram of the the football pitch , note as described earlier the left side of each goal is 33.84 m east of the origin is 7.32 metres in length .

William ,stationed at the origin has the ball is attempting to hit it directly the goal, this is before the game starts so there are no defenders from Schmidders Team

Assume that William can dribble the ball across the edge at a speed of 3 ms^{-1} and if he kicks the ball across the pitch it will travel at a constant speed of 2 ms^{-1}

- a.** If he kicks the ball directly to the middle of the goal, **state** the distance the ball has travelled and the time to the nearest second it takes for the ball to reach it 1 Mark

If William travels initially a distance of a metres in the positive vertical direction up the edge and then kicks the ball to the midpoint of the goal

b.

- i.** State the distance the ball travels in terms of a 1 Mark

- ii.** State the time it takes for the ball travels in terms of a 1 Mark

- iii.** Hence or otherwise, find the the co-ordinate must kick the ball into the field from such that it reaches the ball reaches the midpoint of the north goal the fastest 3 Marks

William is testing what will happen if he changes the speed the ball travels across the football pitch .

Let the speed of the ball across the field be v metres

c.

i. For what value(s) of v will he be able to kick the ball fastest if he kicks it off to the pitch at the halfway line (1 Mark)

ii. For what value(s) of v will the best decision to kick it directly (2 Marks)

When the game begins he suddenly changes his strategy to trick the opponents

He travels directly into the field with the ball and travels to the halfway line, then travels again to the north goal

Schmiddler understands this however and pours a solution of Chemical X, after the halfway line slowing him down to $v \text{ ms}^{-1}$ where $v < 5$

Let p be the horizontal distance across the pitch travels up to the halfway .

d.

i. If $v = 1$ Determine the fastest time it can take for William to put the ball in the midpoint of the goal (3 Marks)

Mrs Sings now the goalie of *Schmiddler's* team , sways side to side at a constant speed, it takes 5 seconds her to reach one side of the goal and she starts from the left side

ii. Assuming no other defenders intercept the ball from William except for *Mrs. Sings*

State all values of v of which she can intercept the ball from reaching the north goal ,
assuming William kicks it in the fastest time possible
(3 Marks)

Let θ_1 be the angle subtended by his path up to the halfway line to the horizontal axis
And Let θ_2 be the angle subtended by the path of the ball from the halfway line to the
midpoint of the north goal

e. State for what value of v is the ratio

$$\frac{\cos(\theta_2)}{\cos(\theta_1)} = \frac{6}{7}$$

Imagine if v sapped out of Williams Energy v^2 per second , and he had a maximum energy
balance of 500 J

f. State the maximum value of p (3 Marks)

As the game was going on , William was subbed out at half time ($t=45$ minutes)

Mr. Kermond and William come to a strange area hopefully to find SAC I, indeed they are in luck as this happen . They traverse to china to the Rainy technologies building

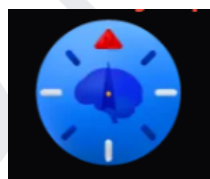
The troupe emerged at a strange horn-like entrance in the region. It was amongst a brazen sky, and amidst the spectral radiance of the stars- accompanied by a candid cacophony of a hummingbirds' harmony . The region was bathed by serenity .

The waves that came seemed be of an oscillatory nature, but neither William nor Mr. Kermond could identify a pure oscillation, it is as if the oscillation could be scaled , by some extraneous parameter. What did this means he pondered to himself

The gang sauntered through the futuristic landscape, it appeared rather clean and a placid marble floor shined a reflection of themselves that seemed purer than they truly were, really it was rather interesting as it exalted their consciousness in a way they could not see.

The enigmatic doors hissed as they opened. Mr. Kermond's and William's faculties were excited-like an electron elevated to a valence shell as they braved the entrance before them, was SAC I located here or was this an elaborate trap by the Schmiddler?

What if- it was an amalgamation of both indeed?



Question 7 (7 Marks)

MISSION STATUS: ONGOING

TIME: 12:00 PM (GMT+8)

LOCATION: 雨天 Technologies Office , 北京 中国

The Rainy technologies corporation is a key player in the Chinese stock-market.

Mr. Kermond and William were led to a random IP address location in China, in the building where where the CEO of Rainy technologies. Mrs. Xia was speaking to a crowd at an auction in the enigmatic conference hall

“我们公司已将这篇论文卖给了出价最高者。这是一篇来自澳大利亚的数学论文。” -
She spoke aloud to the presence of unknown bidders who came from all over the world.

“That’s SAC I” - Mr. Kermond vociferated

“We have to stop her” - William whispered from the air ventilation system

A television screen shows the JKSS Cup Final being broadcasted live internationally, the scores of The *JKSS* team and the *Schmiddler* team are related to two equations

$$L1: k * t - 8y = 6 + 7m \text{ (JKSS)}$$

and

$$L2: 6t + (k - 9)y = 50 \text{ (SCHMIDDLER)}$$

Where $k > 9$ and $m > 0$

t is the time of game in minutes and y is the score of each respective team, the score increase only when the output value exceeds and integer

a.

- i. If $L1$ was infinitely tangent to all local maximums of the curve $3t \sin(t)$, state the values of k and m 2 Marks

- ii. Can $L2$ ever be infinitely tangential to all local minimums of $3t \sin(t)$? justify your answer

(2 Marks)

b. Determine the relation between k and m such that The JKSS Team will win the Finals (1 Marks)

c. What is relation between the two variables II The JKSS team always be ahead of Schmiddler's Team for the whole game where $m > -6/7$ (2 Marks)

The graph of $L1$ is dilated by a factor of g from the horizontal axis

d. What is the minimum value of g such that it is possible for $L1$ and $L2$ to never intersect (2 Marks)

The intensity of the desire to bet on JKSS's victory in the bidders can be modelled by

$$f(x) = k(x^2 - 2x - 3)$$

Where f is an intensity level from -0.477 to 1

Where x is variable the depends on the time of game such that

$$x(t) = \sqrt{2t + 3} - \ln(t^2 + t) \quad t \in [0, 90]$$

e.

i. State $\text{dom}(f)$ in terms of k

2 Marks

ii. State the value of k

1 Mark

Question 7 (10 Marks)

MISSION STATUS:ONGOING

TIME:10:00 PM (GMT+8)

LOCATION: XIA TECHNOLOGIES BASEMENT

Mrs. Xia tries to outwit the gang using her own mathematical acumen , They fight rigorously but realize they can't beat her without the help of **technology** . She shoots a bullet at the gang that follows the path of of a function whose equation cannot be analytically solved.

The only way to stop the bullet is to use numerical methods to approximate the value of π

William has prepared two UDF's on Mathematica to help solve the root:

Shown is the pseudocode for each one

Inputs: $f(x)$, #a function of x ,a real number
a, # the lower interval endpoint
b, # [CORRUPTED]
max# [CORRUPTED]

Define bisection($f(x)$,a,b,iterations):

if $f(a)*f(b)>0$:

Print ("Invalid Interval")

Elif :

While $i < \text{max}$ Do

Mid $\leftarrow$ [CORRUPTED]

If $f(\text{mid})=0$ Then

Return Mid

Elif $f(a) * f(\text{mid}) < 0$ Then

b $\leftarrow$ mid

Else

[CORRUPTED] $\leftarrow$ Mid

i $\leftarrow$ i+1

Return mid

Inputs : $F(\text{var})$, Function of var

x_0 #[CORRUPTED]

k #[CORRUPTED]

Difference #Difference between iterations

$df(\text{var}) \rightarrow$ derivative of $F(\text{var})$ with respect to var

Define Newton($F(\text{var}), \text{var}, x_0, k$, difference) :

$i \leftarrow$ #[CORRUPTED]

$\text{prev_x} \leftarrow x_0$

While $i <$ #[CORRUPTED] Do:

$\text{next_x} \leftarrow \text{prev_x} - (F(\text{prev_x})/df(\text{prev_x}))$

If $\text{Abs}[\text{next_x} - \text{prev_x}] < \text{difference}$ Then:

Return next_x

Else if :

$\text{prev_x} \leftarrow \text{next_x}$

$i \leftarrow$ #[CORRUPTED]

EndWhile

a. The script has been corrupted by hackers who work for Rainy Technologies- Fill all entries with [CORRUPTED] with the correct statement (2 Marks)

The bullet's path can be modelled as the **inverse** function of

$$f(x) = x^5 + x$$

b. One of William's UDF's is more accurate than the other. Use the most accurate one to find the x-co-ordinate of which bullet is **stationary** - give answers to four decimal places , justify which one should be used with proper reason . (3 Marks)

c. **Write** a piece of pseudocode that measures the **error %** Between the two approximations for the intercepts of the inverse function of $f: [k, \infty) \rightarrow \mathbb{R}, f(t) = 3te^{-t} - 3$ where k is the minimum value such that inverse is defined

[WRITE PSEUDOCODE IN THIS SPACE]

William believes an approximation method is the key to fighting her transcendental equations Mr. Kermond informs him of a snippet of pseudocode

d. The **taylor** expansion is a method to approximate a function as a sum of finite polynomials consider a snippet of pseudocode for Taylor series shown:

Inputs : $F(x)$ # function of x
 x_0 # center point of expansion
 k # number of terms
 x # evaluation point

Define TaylorSeries($F(x), x_0, k, x$)

$dF(x) \leftarrow F(x)$

 factorial $\leftarrow 1$

 result $\leftarrow 0$

$n \leftarrow 0$

While $n < k$ Do:

 If $n = 0$ Then

 factorial $\leftarrow 1$

 Else

 factorial $\leftarrow \text{factorial} * n$

 coeff $\leftarrow dF(x_0) / \text{factorial}$

 result $\leftarrow \text{result} + \text{coeff} * (x - x_0)^n$

$dF(x) \leftarrow d(dF(x))$ # advance to next derivative

$n \leftarrow n + 1$

EndWhile

Return result

e. Use this pseudocode to approximate the value of π to 10 decimal places (3 Marks)

They find when the bullet's gradient is approximately equal to $3.141592... \text{ ms}^{-1}$, She has a lapse in consciousness and attack at that time

Mr. Kermond tries to fight Ms. Xia and successfully grabs SAC I

Suddenly she disappears- she was a hyperrealistic holographic

"Huh" they scream

They run back to JMSS Cup Final stadiums and found the Schmiddler has arrived there

The Schmiddler looms in a manner so ominous - infidelity clouds their minds

Welcome to your doom friends

SAC CONTINUES ON THE NEXT PAGE

Question 8 (12 Marks)

MISSION STATUS:[CORRUPTED]

TIME:[CORRUPTED]

LOCATION: [LOG-TERMINATED]

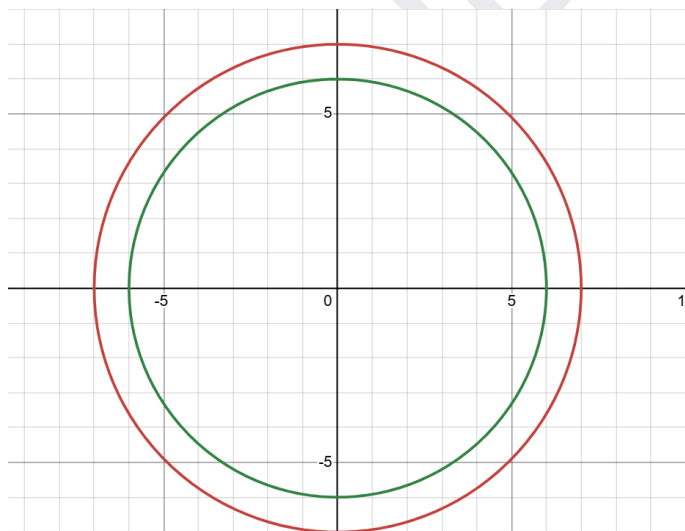
"I've had enough of these games" - Mr. Kermond bellowed

"Why so - because you always lose?" - The schmiddler sneered

"Ah! What's going on guys" - William cries

The Schmiddler clouds William's imaginations with his largest mathematical fears, but he reacts rather interestingly with his newfound relationship with chemical X

The stadium transforms into an annulus structure . William and Mr. Kermond a stuck in a circle of a 6 meter radius relative to an origin . The Schmiddler is situated at a point of the edge of an outer circle of 7 meter radius as shown:



Using the same technology used to make ALBD AI The schmiddler uses a wall to encase the room that has to intersect tangent to a certain point of the outer the circle . The line must intersect the vertical height of the origin 14 metres west .

- a. Find the area bounded by the axes , the wall and the outer circle correct to 2 decimal places
(3 Marks)

- b. Schmiddler now moves to the origin and kicks the ball and it follows a path modelled by $p(x) = k(\sqrt{x} - x)$, $0 < x < k$

Determine the value of k such that it touches the circle and bounces down (2 Marks)

- c. Consider the case in which the ball touches the outer circle twice , a triangle , between between the local maxima of p and the

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